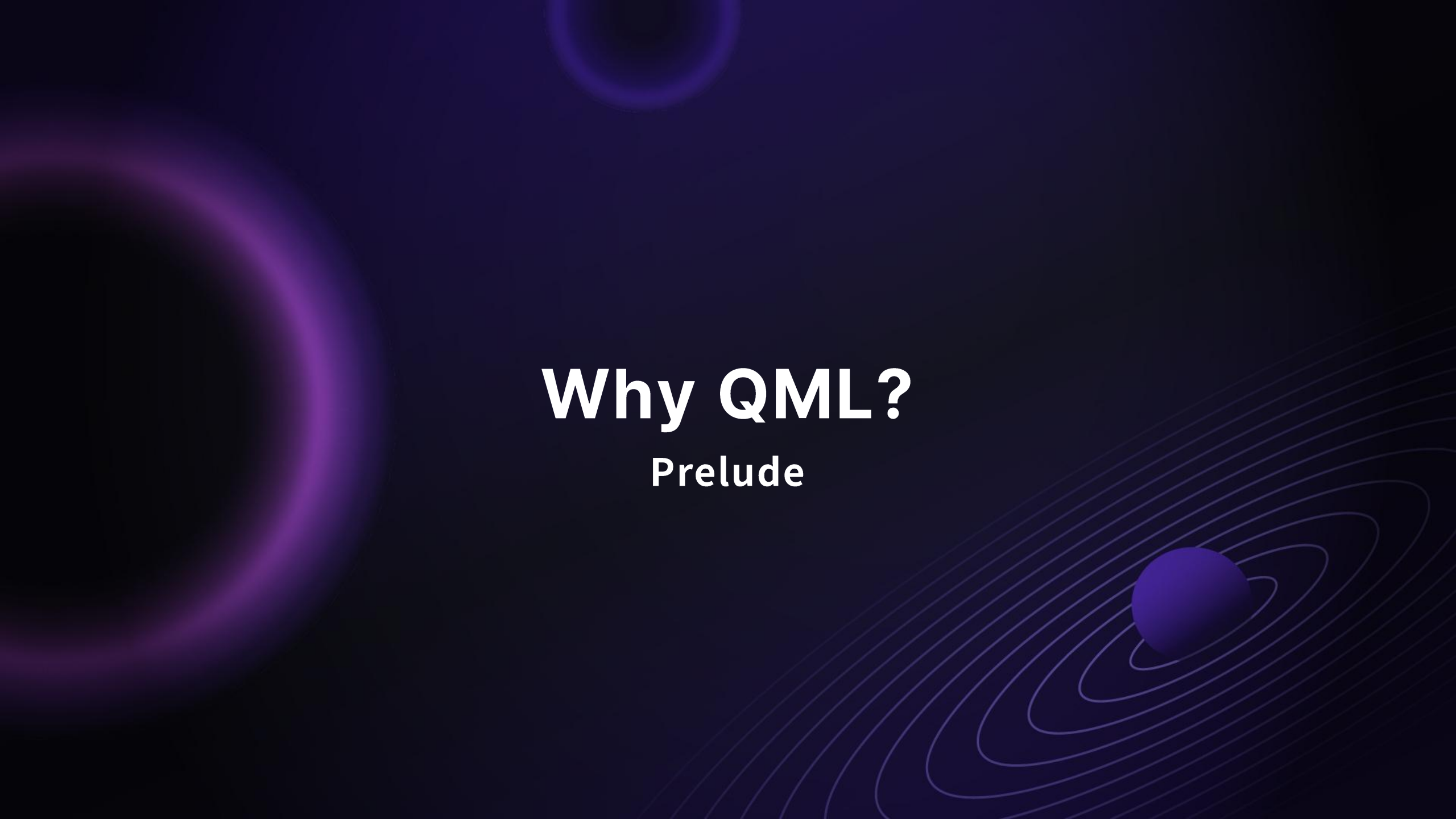


A Tutorial Example for: **The 3rd Global Quantum AI Competition** **1st Challenge**

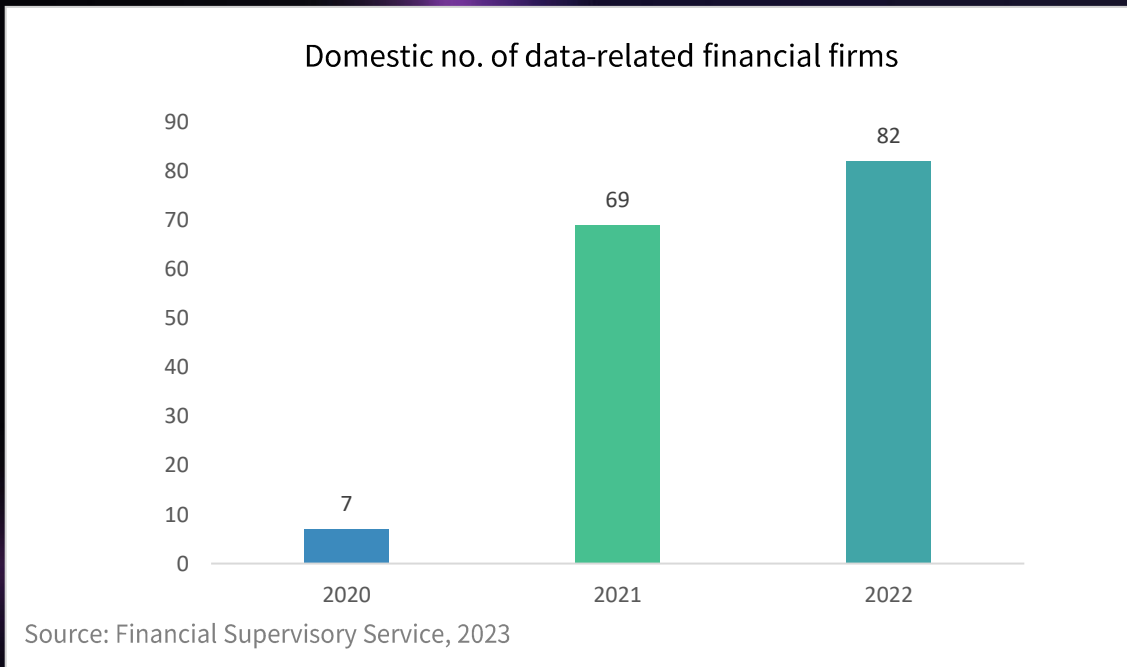
Why QML?

Prelude

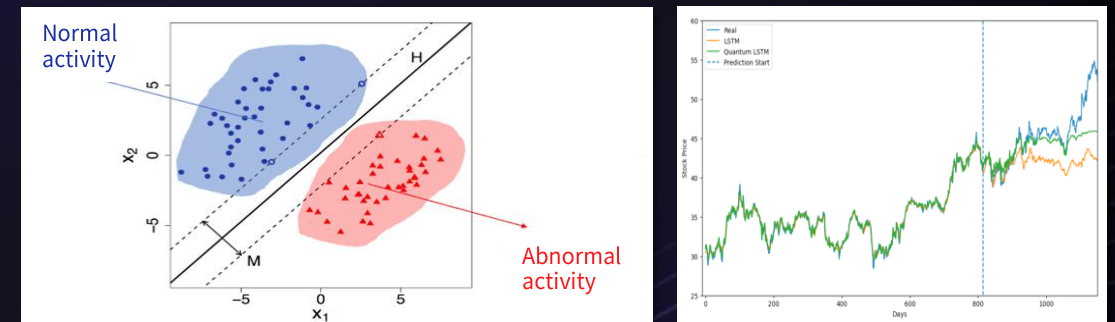


QML & Finance: A Real-World Classification Example

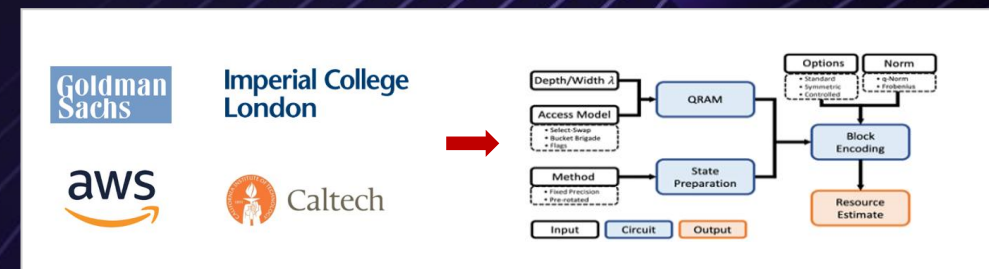
- Increase in financial data usage & no. of data-based firms
- Finance is a representative field of real-world tabular data. Payment histories, transaction records, and customer behavior often contain complex hidden patterns.
- **Credit rating** is a useful reference case for understanding QML-based binary classification.



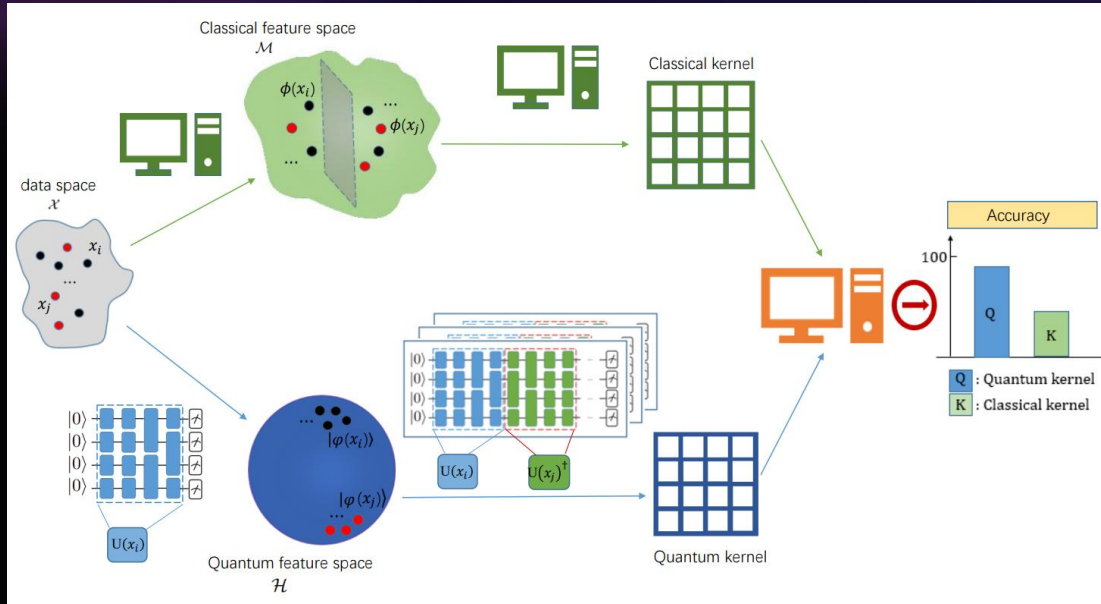
- Quantum Machine Learning (QML)
 - Encodes classical data into quantum circuits.
 - Learns complex, nonlinear patterns through quantum operations.
 - Can be applied to classification tasks such as normal/abnormal activity or delayed/non-delayed payment.



- Reference Use Case
 - QML research with financial data
 - Credit rating / payment-delay prediction using VQC
 - Practical example for QML on real-world tabular data



Why QML?



X. Wang et al., "Towards understanding the power of quantum kernels in the NISQ era," *Quantum* **5**, 531 (2021).

► Classical

Real-world tabular data often contains hidden nonlinear patterns.

Classical models may require complex feature mapping to separate such data effectively.

► Quantum

QML encodes classical input data into quantum states.

Using a limited number of qubits, quantum circuits can represent data in a high-dimensional Hilbert space.

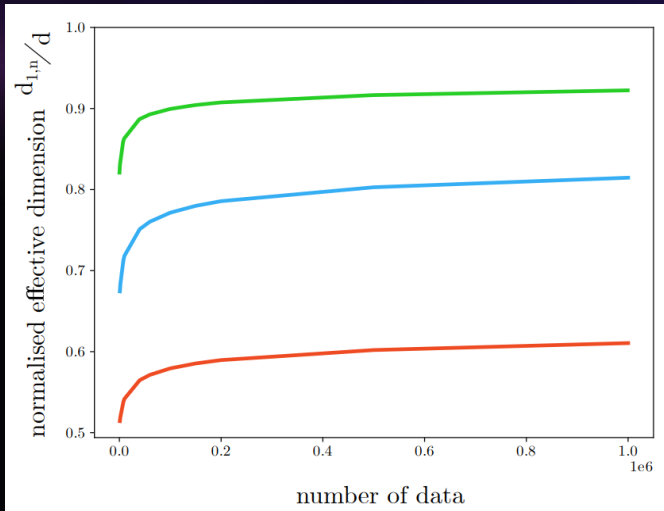
Through rotations and entanglement, quantum circuits can learn decision boundaries for binary classification tasks.

- QML provides a new way to model hidden patterns in tabular data
- Quantum circuits can be used as a practical tool for binary classification

Credit Rating with VQCs

A Worked Example of Binary Classification with QML

The Capabilities of QML



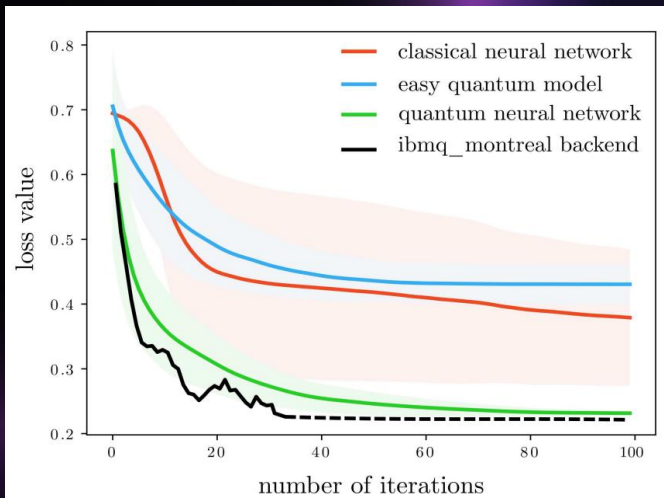
► Effective Dimension

Effective dimension describes how much of a model's capacity is actually used for learning. It is different from simply counting the number of parameters.

In some studies, quantum neural networks show high effective dimension even with a relatively small number of trainable parameters.

This suggests that a compact quantum circuit may still express non-trivial decision boundaries.

For real-world tabular data, such as survey responses or credit-rating records, a small quantum circuit can be used as a meaningful baseline model.



► Convergence Performance

A quantum circuit is trained through an iterative hybrid process.

First, the input data is encoded into the quantum circuit. Then, the circuit is measured to produce a prediction. After that, a classical optimizer updates the trainable angles to reduce the loss.

The loss curve shows how the model gradually improves over training iterations.

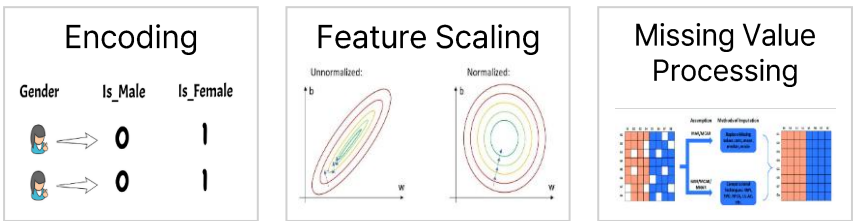
For beginners, the key point is simple: A quantum circuit can be trained in a familiar machine-learning workflow – prediction, loss calculation, parameter update, and evaluation.

The Workflow

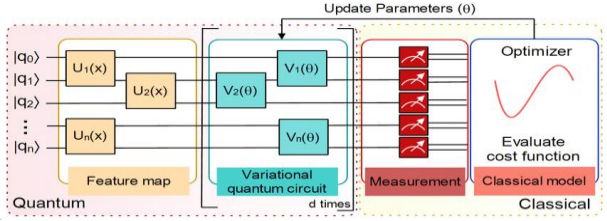
Payment History Data

QML-Based Credit Rating

Input Data Processing

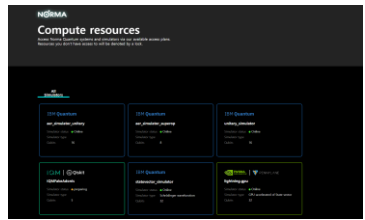


Variational Quantum Classifier



Trainable quantum circuit

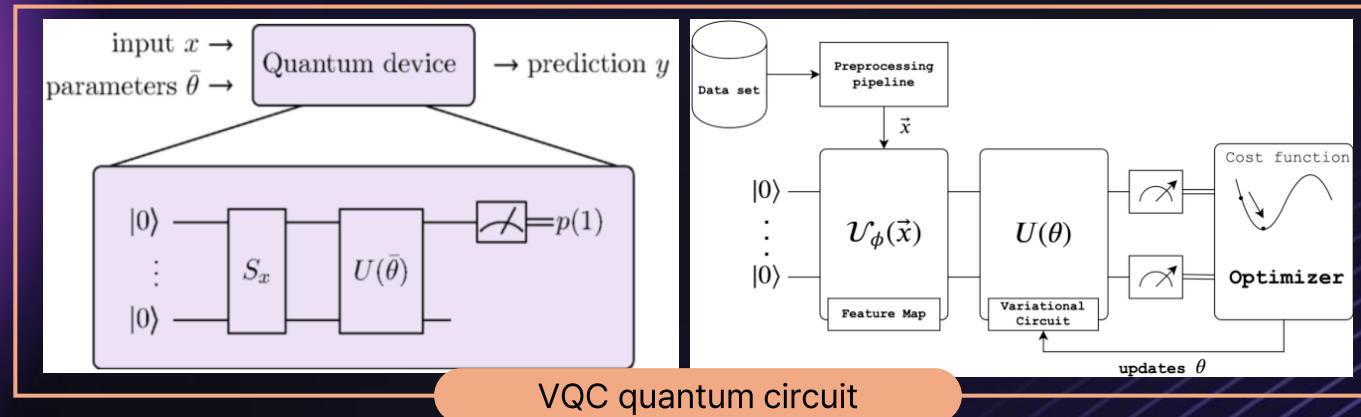
Quantum Emulator



Binary Classification: Predict Delinquent Credit Card Payments

QML → Variational Quantum Classifier (VQC)

- A VQC is a quantum machine learning model for binary classification.
- It consists of two main parts: a data encoding circuit and a trainable variational circuit.
- The encoding circuit maps classical input data into a quantum state, and the variational circuit learns trainable decision boundaries.
- The prediction value is produced by measuring the Z observable of a chosen qubit(s)
- The circuit is optimized through a hybrid workflow: quantum circuit execution + classical parameter update.
- In this reference case, the VQC is used to classify whether a customer is likely to be late for a payment or not.

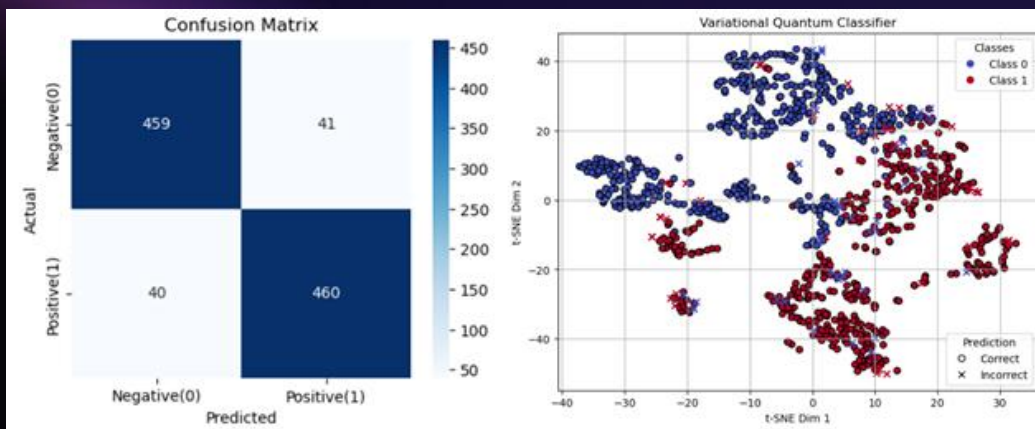


→ **VQC** provides a practical workflow for binary classification on real-world tabular data

→ Input data is encoded into a quantum circuit, optimized through training, and converted into a class prediction by measurement

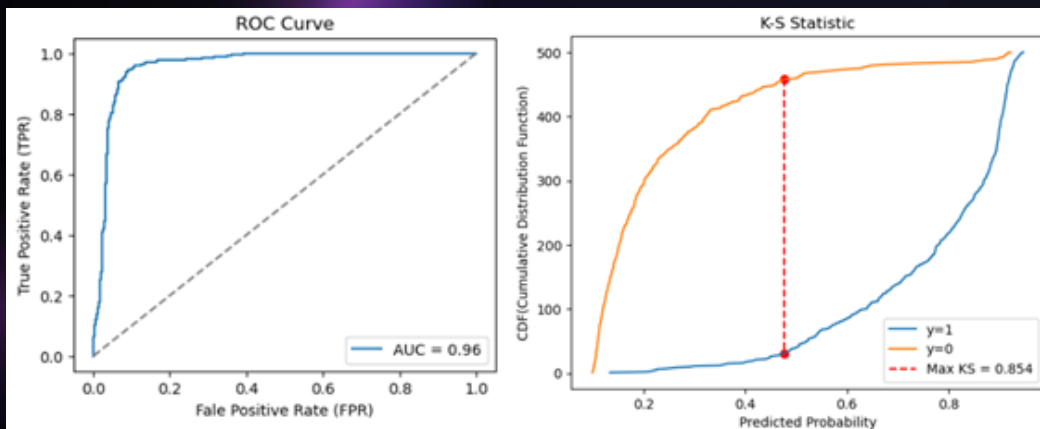
Example Results

► Confusion matrix & visualisation of predicted/actual data



NORMA Quantum AI Team

► ROC curve & KS statistic



NORMA Quantum AI Team

Accuracy
91.9%

AUC 0.96

K-S statistic 85.4%